1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

Mechanistic: In HER2-positive tumors with TLS-like lymphoid aggregates, high tumor-derived CHI3L1 drives a PD-L1–dominant, immunosuppressive state that impairs antigen presentation and T cell priming within TLS, thereby limiting productive anti-tumor immunity and reducing response to HER2-targeted therapy and PD-1/PD-L1 inhibitors. Clinical: Assessing TLS presence together with CHI3L1 and PD-L1 (CD274) status will identify patients who exhibit discordant immune architecture (TLS present but functionally suppressed) and have poorer therapy response; these patients may benefit from combination strategies targeting CHI3L1 or PD-L1 pathways in addition to HER2-targeted therapy. Secondary modifiers (CD8A, mtDNA_content, SUVmax, CPS, MUC1, LINE-1 methylation) will serve as orthogonal readouts to refine risk stratification and mechanistic interpretation, and to predict which patients are most likely to convert benefit with combination regimens.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	CHI3L1, CD274
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CHI3L1: CHI3L1 → PD-L1/inflammatory modulation path, because together they provide a mechanistic route by which tumor-secreted inflammation (CHI3L1) can reprogram checkpoint expression and suppress effective antigen presentation within TLS.

CD274: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CD8A (TIL marker)	Supporting modifier from knowledge graph
Secondary	mtDNA_content	Supporting modifier from knowledge graph

Secondary	SUVmax	Supporting modifier from knowledge graph
Secondary	CPS (PD-L1 Combined Positive Score)	Supporting modifier from knowledge graph
Secondary	MUC1 (CA15-3)	Supporting modifier from knowledge graph
Secondary	LINE-1 methylation	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Retrospective cohort analysis to correlate TLS presence, CHI3L1, PD-L1, and secondary modifiers with therapy response in HER2+ patients treated with HER2-targeted therapy ± immunotherapy.

Prediction 2: Spatial multi-omic studies to assess CHI3L1-driven PD-L1 expression within TLS micro-niches and its impact on T cell activation markers.

Prediction 3: In vitro/in vivo models to test whether CHI3L1 blockade or combined CHI3L1/PD-L1 inhibition improves T cell priming in TLS-like structures.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CD8A (TIL marker): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- mtDNA_content: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.